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DCIODEfTopic

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About the DCIO

[About the DCIO](#)

The UltraFLEX Digital Channel I/O (DCIO) instrument is used for design-for-test (DFT) applications where pins require low speed state control. The DCIO board works with the HSD1000 instrument board to create optimal test system configurations that can handle a large number of low-speed and DC pins. While the HSD1000 instrument board tests high-speed pins, the DCIO uses pattern-controlled 200kHz digital pin electronics and pin parametric measurement units (PPMUs) to test low-speed and DC pins.

The DCIO has 192 channels per instrument board. A test system can have up to 23 DCIO boards, but you must have at least one HSD1000 instrument board to control the DCIO operation. The DCIO also enables you to set up parallel testing and multiple time domains. You can control the PPMUs either with the test system computer or with test patterns. Finally, DCIO has a merged PPMU feature that allows higher current capability.

This section includes the following topics:

- [Features](#)

- [Safety Information](#)

- [Key Terms](#)

- [DCIO Software Licensing Requirements](#)

DCIO information includes the following sections:

- [DCIO Theory of Operation](#)

- [DCIO Programming](#)

- [DCIO Debug Displays](#)

- [DCIO Examples](#)

Related Topics

- [DCIO Specifications](#)

- [DCIO \(VBT syntax\)](#)

- [DCIO DIB Design Guidelines](#)

- [About the HSD1000](#)

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About the DCIO : Features

Features

Each DCIO board contains the following:

- | | |
|---|---|
| • | 192 channels |
| • | Multiple time domains (one time domain per board) |

Pin Electronics

- | | |
|---|--|
| • | Pattern-controlled |
| • | 200kHz data rate |
| • | VIL, VIH, VOH, VOL, VT levels available |
| • | Voltage range from -1V to 6V with a 1mV resolution |
| • | 52K vectors of pattern memory |
| • | PE strobe for high, low, and midband |
| • | Fail capture memory History RAM (4k deep) |

- Synchronization between DCIO and HSD1000 instruments using Instrument Sync-Link (ISL) bus
- Edge alignment of 2.5s 200ns following a vector increment instruction

Computer Controlled Pin Parametric Measurement Units (PPMUs)

- Force current measure voltage mode
- Force voltage measure current mode
- Seven current ranges

Vector Controlled PPMUs

- Force current measure voltage (FIMV) mode
- Force voltage measure current (FVMI) mode
- Pin electronics mode
- Modes programmable with pattern control syntax
- PPMU strobe for high, low, and midband
- Capture measured values at programmed data rate
- Return measured value for high, low, and midband in FIMV and FVMI modes
- Accumulate up to 255 measurements in a pattern burst

Merged PPMU

Up to 32 DCIO channels can be merged together to create a high current PMU. With the maximum merging, a DCIO pin can drive up to 2.048A. For information on the available merged supplies per instrument and maximum merged current, see DCIO Merged PPMU.

For DCIO hardware specifications, see [DCIO Specifications](#).

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About the DCIO : Safety Information

Safety Information

Safe use of test system instruments is important. To ensure that you are using the instrument safely, read and follow the safety information in About Operator and Safety Information.

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About the DCIO : Key Terms

Key Terms

ISL	Instrument Sync-Link
MLT	Module lookup table
MRAM	Per-pin measurement RAM
VMT	Vector module table

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About the DCIO : DCIO Software Licensing Requirements

DCIO Software Licensing Requirements

Operation of the DCIO does not require a software license.
For more information on IG-XL licensing, see [IG-XL Licensing](#) in IG-XL Administration help.

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DCIO Examples

DCIO Examples

This section contains a program example for the DCIO instrument. This IG-XL workbook example shows all IG-XL data, VBT code, and patterns necessary for a complete running program. The following topic is available:

- [DCIO PPMU Test Programming](#)

Related Topics

- [DCIO \(VBT Syntax Reference\)](#)
- [DCIO Programming](#)

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DCIO Examples : DCIO PPMU Test Programming

DCIO PPMU Test Programming

This workbook example shows how to program the DCIO on an HSD1000 loopback DIB to perform the following tests:

- A standard PPMU test using DCIO channels
- A pattern-controlled PPMU test using DCIO channels

This example uses one DCIO board in slot 9 and one HSD1000 board in slot 14.
This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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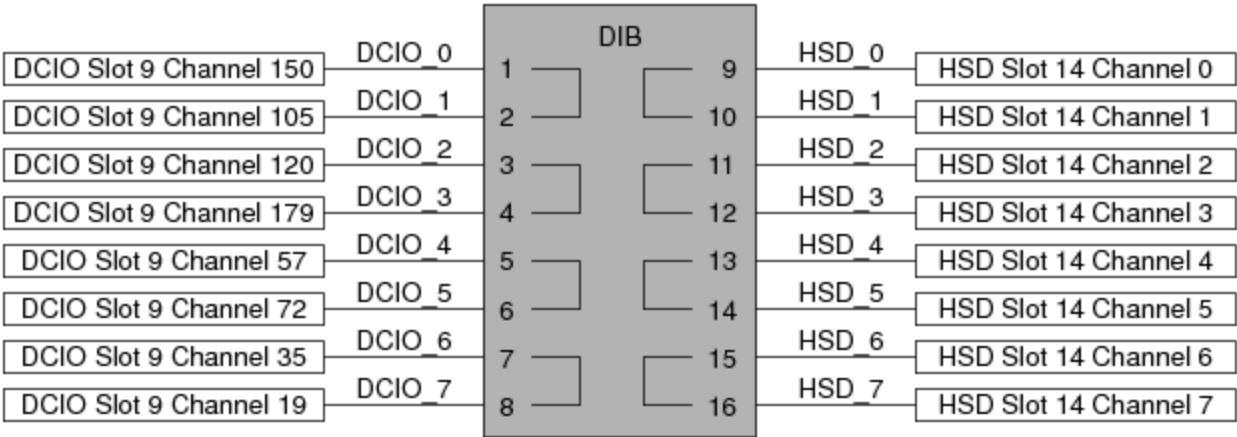
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DCIO Examples : [DCIO PPMU Test Programming](#) : DUT Connections

DUT Connections



Connections for HSD1000 Loopback DIB

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DCIO Examples : [DCIO PPMU Test Programming](#) : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- | | |
|---|---|
| • | Pin Map Sheet |
| • | Channel Map Sheet |
| • | Pin Levels Sheet |
| • | Global Specs Sheet |
| • | DC Specs Sheet |
| • | Time Sets (Basic) Sheet |
| • | Test Instances Sheet |
| • | Pattern Sets Sheet |
| • | Flow Table Sheet |

Pin Map Sheet

Defining the relevant pins:

Pin Map

Group Name	Pin Name	Type
	DCIO_0	I/O
	DCIO_1	I/O
	DCIO_2	I/O
	DCIO_3	I/O
	DCIO_4	I/O
	DCIO_5	I/O
	DCIO_6	I/O
	DCIO_7	I/O
	HSD_0	I/O
	HSD_1	I/O
	HSD_2	I/O
	HSD_3	I/O
	HSD_4	I/O
	HSD_5	I/O
	HSD_6	I/O
	HSD_7	I/O

Defining the relevant pin groups:

Pin Map

Group Name	Pin Name	Type
DCIO_EVEN_PINS	DCIO_0	I/O
DCIO_EVEN_PINS	DCIO_2	
DCIO_EVEN_PINS	DCIO_4	
DCIO_EVEN_PINS	DCIO_6	
DCIO_ODD_PINS	DCIO_1	I/O
DCIO_ODD_PINS	DCIO_3	
DCIO_ODD_PINS	DCIO_5	
DCIO_ODD_PINS	DCIO_7	
HSD_EVEN_PINS	HSD_0	I/O
HSD_EVEN_PINS	HSD_2	
HSD_EVEN_PINS	HSD_4	
HSD_EVEN_PINS	HSD_6	
HSD_ODD_PINS	HSD_1	I/O
HSD_ODD_PINS	HSD_3	
HSD_ODD_PINS	HSD_5	
HSD_ODD_PINS	HSD_7	
ALL_EVEN_PINS	DCIO_EVEN_PINS	I/O
ALL_EVEN_PINS	HSD_EVEN_PINS	
ALL_ODD_PINS	DCIO_ODD_PINS	I/O
ALL_ODD_PINS	HSD_ODD_PINS	
ALL_DCIO	DCIO_EVEN_PINS	I/O
ALL_DCIO	DCIO_ODD_PINS	
ALL_HSD	HSD_EVEN_PINS	I/O
ALL_HSD	HSD_ODD_PINS	

[Channel Map Sheet](#)

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel		Comment
Pin Name	Package Pin	Type	Site 0	
DCIO_0		DCIO	9.ch150	
DCIO_1		DCIO	9.ch105	
DCIO_2		DCIO	9.ch120	
DCIO_3		DCIO	9.ch179	
DCIO_4		DCIO	9.ch57	
DCIO_5		DCIO	9.ch72	
DCIO_6		DCIO	9.ch35	
DCIO_7		DCIO	9.ch19	
HSD_0		I/O	14.ch0	
HSD_1		I/O	14.ch1	
HSD_2		I/O	14.ch2	
HSD_3		I/O	14.ch3	
HSD_4		I/O	14.ch4	
HSD_5		I/O	14.ch5	
HSD_6		I/O	14.ch6	
HSD_7		I/O	14.ch7	

[Pin Levels Sheet](#)

Defined using formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
ALL_DCIO		Vil	=_dcio_vil
ALL_DCIO		Vih	=_dcio_vih
ALL_DCIO		Vol	=_dcio_vol
ALL_DCIO		Voh	=_dcio_voh
ALL_DCIO		Vt	=_dcio_vt
ALL_DCIO		ForceV	=_dcio_forcev
ALL_DCIO		ForceI	=_dcio_forcei
ALL_DCIO		ComparelHi	=_dcio_cmpihi
ALL_DCIO		ComparelLo	=_dcio_cmpilo
ALL_DCIO		CompareVHi	=_dcio_cmpvhi
ALL_DCIO		CompareVLo	=_dcio_cmpvlo
ALL_DCIO		ForceIRange	=_dcio_frcirng
ALL_DCIO		MeasureIRange	=_dcio_measirng
ALL_DCIO		ClampVHi	=_Vch_default
ALL_DCIO		ClampVLo	=_Vcl_default
ALL_DCIO		DriverMode	HiZ
ALL_HSD		Vil	0
ALL_HSD		Vih	5
ALL_HSD		Vol	0.5
ALL_HSD		Voh	4.5
ALL_HSD		Vt	2.5
ALL_HSD		Vcl	=_Vcl_default
ALL_HSD		Vch	=_Vch_default
ALL_HSD		VoutLoTyp	1
ALL_HSD		VoutHiTyp	4
ALL_HSD		DriverMode	Largeswing-VT

Formulas evaluated:

Pin Levels

Pin/Group	Seq.	Parameter	Value
ALL_DCIO		Vil	0
ALL_DCIO		Vih	5
ALL_DCIO		Vol	0.5
ALL_DCIO		Voh	4.5
ALL_DCIO		Vt	2.5
ALL_DCIO		ForceV	5
ALL_DCIO		Forcel	0
ALL_DCIO		ComparelHi	0.01
ALL_DCIO		ComparelLo	-0.01
ALL_DCIO		CompareVHi	4.5
ALL_DCIO		CompareVLo	0.5
ALL_DCIO		ForcelRange	0.002
ALL_DCIO		MeasurelRange	0.032
ALL_DCIO		ClampVHi	6
ALL_DCIO		ClampVLo	-1
ALL_DCIO		DriverMode	HiZ
ALL_HSD		Vil	0
ALL_HSD		Vih	5
ALL_HSD		Vol	0.5
ALL_HSD		Voh	4.5
ALL_HSD		Vt	2.5
ALL_HSD		Vcl	-1
ALL_HSD		Vch	6
ALL_HSD		VoutLoTyp	1
ALL_HSD		VoutHiTyp	4
ALL_HSD		DriverMode	Largeswing-VT

Global Specs Sheet

Global Specs

Symbol	Job	Value	Comment
Vcl_default	-	1.E+00	Detector clamp voltage low
Vch_default		6.E+00	Detector clamp voltage high

DC Specs Sheet

DC Specs



Symbol	Value	Selector		DCIO Specs	DCIO HiZ	DCIO MV 2 5V	DCIO DRV PE
		Name	Val	Typ	Typ	Typ	Typ
dcio_vil	0	DC_Type	Typ	0	0	0	0
dcio_vih	5	DC_Type	Typ	5	5	5	5
dcio_vol	0.5	DC_Type	Typ	0.5	0.5	0.5	0.5
dcio_voh	4.5	DC_Type	Typ	4.5	4.5	4.5	4.5
dcio_vt	2.5	DC_Type	Typ	2.5	2.5	2.5	2.5
dcio_forcev	5	DC_Type	Typ	5	5	5	5
dcio_forcei	0	DC_Type	Typ	0	0	0	0
dcio_cmpihi	0.01	DC_Type	Typ	0.01	0.01	0.01	0.01
dcio_cmpilo	-0.01	DC_Type	Typ	-0.01	-0.01	-0.01	-0.01
dcio_cmpvhi	4.5	DC_Type	Typ	4.5	4.5	2.6	4.5
dcio_cmpvlo	0.5	DC_Type	Typ	0.5	0.5	2.4	0.5
dcio_frcirng	0.032	DC_Type	Typ	0.032	0.002	0.002	0.002
dcio_measirng	0.032	DC_Type	Typ	0.032	0.032	0.032	0.032
dcio_term	VT	DC_Type	Typ	VT	HiZ	VT	VT

Time Sets (Basic) Sheet

Time Sets (Basic)



Timing Mode: Master Timeset Name:

Time Domain:

	Cycle	Pin/Group			Data		Drive			Compare			Edge Resolution	
Time Set	Period	Name	Clock Period	Setup	Src	Fmt	On	Data	Return	Off	Mode	Open	Close	Mode
tset_1	200.E-09	ALL_HSD			PAT	NR	0	25.E-09		200.E-09	Edge	175.E-09		Auto

Test Instances Sheet

Test Instances



Test Name	Test Procedure			DC Specs		AC Specs		Sheet Parameters				
	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
CPU_PPMU	VBT	PinPmu_T	Excel Macro	DCIO_Specs	DC_Type					Levels		
CPU_MVOdd	VBT	Functional_T	Excel Macro	DCIO_Specs	DC_Type			TimeSets		Levels		
FVEven_MVOdd	VBT	Functional_T	Excel Macro	DCIO_MV_2_5V	DC_Type			TimeSets		Levels		
FIEven_MIOdd	VBT	Functional_T	Excel Macro	DCIO_Specs	DC_Type			TimeSets		Levels		
StbMid_All_Modes	VBT	Functional_T	Excel Macro	DCIO_HiZ	DC_Type			TimeSets		Levels		
VBT_PEEven_MVOdd	VBT	Functional_T		DCIO_DRV_PE	DC_Type			TimeSets		Levels		

The Functional_T test instances use the following patterns.

Test Instances

Test Name	Patterns
CPU_MVOdd	.\Patterns\DCIO_Module_PE_CPU_MV.PAT
FVEven_MVOdd	FVEven_MVOdd
FIEven_MIOdd	FIEven_MIOdd
StbMid_All_Modes	StbMid_All_Modes
VBT_PEEven_MVOdd	.\Patterns\DCIO_Module_PEEven_MVOdd.PAT

The CPU_PPMU test instance has the following test-specific parameters and values as shown in the Instance Editor:

Instance Editor [VBT Test 'PinPmu_T']

Instance: CPU_PPMU

Parameters | Timing and Levels | Limits

Name	Value	Comment
Pins	ALL_EVEN_PINS	
MeasureMode	tPPMUForceIMeasureV	
ForceCond1	0	
ForceCond2	0	
ForceIRange	Auto	Valid if forcing I
MeasIRange		Valid if forcing V
VClampLo		V
VClampHi		V
RelayMode	tUnpowered	
SettlingTime	0	s
SamplingTime	0	s
SampleCount	0	
+ Testing Pins Serially		
- Interpose Functions		
StartOfBodyF	SetLevelsFVOdd	
StartOfBodyFArgs	5	
PrePatF		
PrePatFArgs		
PreTestF		
PreTestFArgs		
PostTestF		
PostTestFArgs		
PostPatF		
PostPatFArgs		
EndOfBodyF		
EndOfBodyFArgs		
+ Precondition Pattern		
+ Hold State Pattern		
+ Pin States		

Pattern Sets Sheet

Pattern Sets

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Pattern Set	TD Group	Time Domain	File/Group Name	Burst	Start Label	Stop Label
PEEven_MVOdd			Patterns/NON_DCIO_Module_NoHalt.PAT	yes		
PEEven_MVOdd			Patterns/DCIO_Module_PEEven_MVOdd.PAT	yes		
FIEven_MIOdd			Patterns/NON_DCIO_Module_NoHalt.PAT	yes		
FIEven_MIOdd			Patterns/DCIO_Module_FIEven_MIOdd.PAT	yes		
StbMid_All_Modes			Patterns/NON_DCIO_Module_NoHalt.PAT	yes		
StbMid_All_Modes			Patterns/DCIO_Module_StbMid_All_Modes.PAT	yes		
FVEven_MVOdd			Patterns/NON_DCIO_Module_NoHalt.PAT	yes		
FVEven_MVOdd			Patterns/DCIO_Module_FVEven_MVOdd.PAT	yes		

Flow Table Sheet

Flow Table

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		Gate			Command				Limits		Bin Number		Sort Number		Result
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Pass	Fail	Pass	Fail	
	CPU				Test	CPU_PPMU		100			1	1001	1	1001	
					Use-Limit	CPU_PPMU			4.9	5.1	1	1001	1	1001	
	MV				Test	CPU_MVOdd		200			1	1002	1	1002	
	MV				Test	FVEven_MVOdd		300			1	1003	1	1003	
	MI				Test	FIEven_MIOdd		400			1	1004	1	1004	
	ALLMODES				Test	StbMid_All_Modes		500			1	1005	1	1005	
	MULTI				Test	VBT_PEEven_MVOdd		600			1	1000	1	1000	
					stop										

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[DCIO Examples](#) : [DCIO PPMU Test Programming](#) : Patterns

Patterns

[DCIO_Module_FVEven_MVOdd.PAT](#) Pattern File

DCIO_Module_FVEven_MVOdd.PAT - DCIO_FVEven_MVOdd																				
				-																
				D C I O	D C I O	D C I O	D C I O	D C I O	D C I O	D C I O	D C I O		H S D	H S D	H S D	H S D	H S D	H S D	H S D	
TSet	Command	Label	Vector/ Cycle	0 C C	1 C C	2 C C	3 C C	4 C C	5 C C	6 C C	7 C C		0 D	1 D	2 D	3 D	4 D	5 D	6 D	
tset_1			22										0	0	0	0	0	0	0	
tset_1			23										0	0	0	0	0	0	0	
tset_1			24										0	0	0	0	0	0	0	
tset_1			25										0	0	0	0	0	0	0	
tset_1			26										0	0	0	0	0	0	0	
tset_1			27										0	0	0	0	0	0	0	
tset_1			28										0	0	0	0	0	0	0	
tset_1			29										0	0	0	0	0	0	0	
tset_1			30										0	0	0	0	0	0	0	
tset_1			31										0	0	0	0	0	0	0	
tset_1			32	K	C	K	C	K	C	K	C		0	0	0	0	0	0	0	
tset_1	repeat 500		33										0	0	0	0	0	0	0	
tset_1			34	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		35										0	0	0	0	0	0	0	
tset_1			36	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		37										0	0	0	0	0	0	0	
tset_1			38	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		39										0	0	0	0	0	0	0	
tset_1			40	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		41										0	0	0	0	0	0	0	
tset_1			42	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		43										0	0	0	0	0	0	0	
tset_1			44	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		45										0	0	0	0	0	0	0	
tset_1			46	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		47										0	0	0	0	0	0	0	
tset_1			48	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		49										0	0	0	0	0	0	0	
tset_1			50	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		51										0	0	0	0	0	0	0	
tset_1			52	K	M	K	M	K	M	K	M		0	0	0	0	0	0	0	
tset_1	repeat 25		53										0	0	0	0	0	0	0	

DCIO_Module_FIEven_MIOdd.PAT Pattern File

DCIO_Module_FIEven_MIOdd.PAT - DCIO_FIEven_MIOdd												
TSet	Command	Label	Vector/ Cycle	-								
				D	D	D	D	D	D	D	D	
				C	C	C	C	C	C	C	C	
				I	I	I	I	I	I	I	I	
				O	O	O	O	O	O	O	O	
				0	1	2	3	4	5	6	7	
				(	(	(	(	(	(	(	(	
				D	D	D	D	D	D	D	D	
				C	C	C	C	C	C	C	C	
				H	S	S	S	S	S	S	S	
				D	D	D	D	D	D	D	D	
				0	1	2	3	4	5	6	7	
tset_1			0							0	0	0
tset_1			1							0	0	0
tset_1			2							0	0	0
tset_1			3							0	0	0
tset_1			4							0	0	0
tset_1			5							0	0	0
tset_1			6	C	K	C	K	C	K	C	K	0
tset_1	repeat 500		7							0	0	0
tset_1			8	C	L	C	M	C	H	C	L	0
tset_1	repeat 25		9							0	0	0

DCIO_Module_PEEven_MVOdd.PAT Pattern File

DCIO_Module_PEEven_MVOdd.PAT - DCIO_PEEven_MVOdd												
TSet	Command	Label	Vector/ Cycle	-								
				D	D	D	D	D	D	D	D	
				C	C	C	C	C	C	C	C	
				I	I	I	I	I	I	I	I	
				O	O	O	O	O	O	O	O	
				0	1	2	3	4	5	6	7	
				(	(	(	(	(	(	(	(	
				D	D	D	D	D	D	D	D	
				C	C	C	C	C	C	C	C	
				H	S	S	S	S	S	S	S	
				D	D	D	D	D	D	D	D	
				0	1	2	3	4	5	6	7	
tset_1			0							0	0	0
tset_1			1							0	0	0
tset_1			2							0	0	0
tset_1			3							0	0	0
tset_1			4							0	0	0
tset_1			5	1	H	1	H	1	H	1	H	0
tset_1	repeat 25		6							0	0	0
tset_1			7	0	L	0	L	0	L	0	L	0
tset_1	repeat 25		8							0	0	0
tset_1			9							0	0	0
tset_1			10							0	0	0
tset_1			11							0	0	0
tset_1			12	1	H	1	H	1	H	1	H	0
tset_1	repeat 25		13							0	0	0
tset_1			14	0	L	0	L	0	L	0	L	0
tset_1	repeat 25		15							0	0	0

DCIO_Module_StbMid_All_Modes.atp Pattern File

DCIO_Module_StbMid_All_Modes.PAT - DCIO_Stb_Mid																
TSet	Command	Label	Vector/ Cycle	-							H	H	H	H	H	H
				D	D	D	D	D	D	D						
				C	C	C	C	C	C	C						
				I	I	I	I	I	I	I						
				O	O	O	O	O	O	O	S	S	S	S	S	S
				0	1	2	3	4	5	6						
				(	(	(	(	(	(	(						
				D	D	D	D	D	D	D						
				C	C	C	C	C	C	C	0	1	2	3	4	5
tset_1			65	1	M	1	M	1	M	1	M	0	0	0	0	0
tset_1	repeat 500		66									0	0	0	0	0
tset_1			67	0	L	0	L	0	L	0	L	0	0	0	0	0
tset_1	repeat 25		68									0	0	0	0	0
tset_1			69	0	C	0	C	0	C	0	C	0	0	0	0	0
tset_1	repeat 500		70									0	0	0	0	0
tset_1			71	1	M	1	M	1	M	1	M	0	0	0	0	0
tset_1	repeat 25		72									0	0	0	0	0
tset_1			73	0	L	0	L	0	L	0	L	0	0	0	0	0
tset_1	repeat 25		74									0	0	0	0	0
tset_1			75	C	0	C	0	C	0	C	0	0	0	0	0	0
tset_1	repeat 500		76									0	0	0	0	0
tset_1			77	M	1	M	1	M	1	M	1	0	0	0	0	0
tset_1	repeat 25		78									0	0	0	0	0
tset_1			79	L	0	L	0	L	0	L	0	0	0	0	0	0
tset_1	repeat 25		80									0	0	0	0	0
tset_1			81	C	K	C	K	C	K	C	K	0	0	0	0	0

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DCIO Examples : [DCIO PPMU Test Programming](#) : VBT Code

VBT Code

' Programs the HRAM.

```
Function HRAM_Capture_All(Argc As Long, argv() As String)
```

```
TheHdw.DCIO.HRAM.SetTrigger trigFirst, False, 3
```

```
TheHdw.DCIO.HRAM.CaptureType = captAll
```

```
TheHdw.DCIO.Patgen.Events.SetCycleCount True, 4, tlCycleTypeModule
```

```
End Function
```

' Set the MRAM to accumulate multiple measurements.

```
Public Function AccumulateDCIOPPMU(Argc As Long, argv() As String) As Long
```

```
    TheHdw.DCIO.Pins(argv(0)).MRAM.Accumulate = tlOn
```

```
End Function
```

' Datalog the measured values after a DCIO pattern-controlled PPMU test.

```
Public Function DatalogAll(Argc As Long, argv() As String) As Long
```

```
    TheHdw.DCIO.Pins(argv(0)).MRAM.test (tlDCIOMramTestAll)
```

```
    TheHdw.DCIO.Pins(argv(0)).MRAM.Accumulate = tlOff
```

```
End Function
```

' All of the following functions modify DCIO pin levels.

```
Public Function SetLevelsFIEven(Argc As Long, argv() As String) As Long
```

```
TheHdw.DCIO.Pins("ALL_DCIO").Levels.DriverMode = tlDCIODriverModeVt
```

```
TheHdw.DCIO.Pins("DCIO_0,DCIO_2,DCIO_4,DCIO_6").Mode = tlDCIOModeFIMV
```

```
TheHdw.DCIO.Pins("DCIO_1,DCIO_3,DCIO_5,DCIO_7").Mode = tlDCIOModeFVMI
```

```
TheHdw.DCIO.Pins("DCIO_0,DCIO_2,DCIO_4,DCIO_6").StartState = tlDCIOStartStateForcel
```

```
TheHdw.DCIO.Pins("DCIO_1,DCIO_3,DCIO_5,DCIO_7").StartState = tIDCIOStartStateForceV
TheHdw.DCIO.Pins("DCIO_0").Levels.Value(tIDCIOLevelPMUFI) = 0.015
TheHdw.DCIO.Pins("DCIO_2").Levels.Value(tIDCIOLevelPMUFI) = -0.001
TheHdw.DCIO.Pins("DCIO_4").Levels.Value(tIDCIOLevelPMUFI) = -0.015
TheHdw.DCIO.Pins("DCIO_6").Levels.Value(tIDCIOLevelPMUFI) = 0.015
End Function
```

```
Public Function SetLevelsFIOdd(Argc As Long, argv() As String) As Long
TheHdw.DCIO.Pins("ALL_DCIO").Levels.DriverMode = tIDCIODriverModeVt
TheHdw.DCIO.Pins("DCIO_0,DCIO_2,DCIO_4,DCIO_6").Mode = tIDCIOModeFVMI
TheHdw.DCIO.Pins("DCIO_1,DCIO_3,DCIO_5,DCIO_7").Mode = tIDCIOModeFIMV
TheHdw.DCIO.Pins("DCIO_0,DCIO_2,DCIO_4,DCIO_6").StartState = tIDCIOStartStateForceV
TheHdw.DCIO.Pins("DCIO_1,DCIO_3,DCIO_5,DCIO_7").StartState = tIDCIOStartStateForcel
TheHdw.DCIO.Pins("DCIO_1").Levels.Value(tIDCIOLevelPMUFI) = 0.015
TheHdw.DCIO.Pins("DCIO_3").Levels.Value(tIDCIOLevelPMUFI) = -0.001
TheHdw.DCIO.Pins("DCIO_5").Levels.Value(tIDCIOLevelPMUFI) = -0.015
TheHdw.DCIO.Pins("DCIO_7").Levels.Value(tIDCIOLevelPMUFI) = 0.015
End Function
```

```
Public Function SetLevelsStbMid(Argc As Long, argv() As String) As Long
TheHdw.DCIO.Pins("ALL_DCIO").Mode = tIDCIOModePE
TheHdw.DCIO.Pins("ALL_DCIO").Levels.Value(tIDCIOLevelPEVih) = 2.5
TheHdw.DCIO.Pins("ALL_DCIO").Levels.Value(tIDCIOLevelPMUFI) = 0.001
TheHdw.DCIO.Pins("ALL_DCIO").StartState = tIDCIOStartStateLow
End Function
```

```
Public Function SetLevelsPEFI(Argc As Long, argv() As String) As Long
TheHdw.DCIO.Pins("ALL_DCIO").Levels.DriverMode = tIDCIODriverModeVt
TheHdw.DCIO.Pins("ALL_DCIO").StartState = tIDCIOStartStateLow
TheHdw.DCIO.Pins("ALL_DCIO").Levels.Value(tIDCIOLevelFIRange) = 0.002
TheHdw.DCIO.Pins("ALL_DCIO").Levels.Value(tIDCIOLevelPMUFI) = 0.001
End Function
```

```
Public Function SetLevelsPEEven(Argc As Long, argv() As String) As Long
TheHdw.DCIO.Pins("ALL_DCIO").Levels.DriverMode = tIDCIODriverModeHiZ
End Function
```

```
Public Function SetLevelsFVEven(Argc As Long, argv() As String) As Long
```

Dim ForceVal As Double

If Argc = 1 Then ForceVal = CDbI(argv(0))

TheHdw.DCIO.Pins("DCIO_EVEN_PINS").Mode = tIDCIOModeFVMI

TheHdw.DCIO.Pins("DCIO_EVEN_PINS").StartState = tIDCIOStartStateForceV

If Argc = 1 Then

 TheHdw.DCIO.Pins("DCIO_EVEN_PINS").Levels.Value(tIDCIOLevelPMUFV) = ForceVal

 Call TheHdw.PPMU.Pins("HSD_EVEN_PINS").ForceV(ForceVal, 0.05)

Else

 TheHdw.DCIO.Pins("DCIO_0").Levels.Value(tIDCIOLevelPMUFV) = 0

 TheHdw.DCIO.Pins("DCIO_2").Levels.Value(tIDCIOLevelPMUFV) = 2.5

 TheHdw.DCIO.Pins("DCIO_4").Levels.Value(tIDCIOLevelPMUFV) = 5#

 TheHdw.DCIO.Pins("DCIO_6").Levels.Value(tIDCIOLevelPMUFV) = 0

End If

End Function

Public Function SetLevelsFVOdd(Argc As Long, argv() As String) As Long

Dim ForceVal As Double

If Argc = 1 Then ForceVal = CDbI(argv(0))

' TheHdw.DCIO.Pins("DCIO_EVEN_PINS").Mode = tIDCIOModeFIMV

' TheHdw.DCIO.Pins("DCIO_EVEN_PINS").StartState = tIDCIOStartStateForcel

TheHdw.DCIO.Pins("DCIO_ODD_PINS").Mode = tIDCIOModeFVMI

TheHdw.DCIO.Pins("DCIO_ODD_PINS").StartState = tIDCIOStartStateForceV

If Argc = 1 Then

 TheHdw.DCIO.Pins("DCIO_ODD_PINS").Levels.Value(tIDCIOLevelPMUFV) = ForceVal

 TheHdw.Digital.Pins("HSD_ODD_PINS").Disconnect

 Call TheHdw.PPMU.Pins("HSD_ODD_PINS").ForceV(ForceVal, 0.05)

 TheHdw.PPMU.Pins("HSD_ODD_PINS").Gate = tIOn

 TheHdw.PPMU.Pins("HSD_ODD_PINS").Connect

Else

 TheHdw.DCIO.Pins("DCIO_1").Levels.Value(tIDCIOLevelPMUFV) = 0

 TheHdw.DCIO.Pins("DCIO_3").Levels.Value(tIDCIOLevelPMUFV) = 2.5

```
TheHdw.DCIO.Pins("DCIO_5").Levels.Value(tIDCIOLevelPMUFV) = 5#  
TheHdw.DCIO.Pins("DCIO_7").Levels.Value(tIDCIOLevelPMUFV) = 0  
Call TheHdw.PPMU.Pins("HSD_ODD_PINS").ForceV(5, 0.05)  
End If  
End Function
```


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DCIO Debug Displays

DCIO Debug Displays

This section describes the Test Debug Environment (TDE) debug displays used by the DCIO instrument:

- [DCIO Channel Debug Display](#)
- [PPMU Debug Display and DCIO Pins](#)

While both of these debug displays control the same DCIO hardware, which display you use depends on the DCIO operation mode.

- [To debug DCIO channels in Pin Electronics mode, use the DCIO Channel debug display.](#)
- [To debug DCIO channels in PPMU mode \(force current/measure voltage or force voltage/measure current\), use the PPMU debug display.](#)

☑ Note:

Due to the interactive nature of the debug displays, you cannot debug pattern-controlled operation from the DCIO debug displays. Use the PatternTools to debug DCIO pattern-controlled operation.

This section describes features specific to the DCIO Debug Display. For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)

- [Producing Code from Debug Displays](#)

See also:

- [About the DCIO](#)
- [DCIO Theory of Operation](#)
- [DCIO Programming](#)
- [DCIO Examples](#)

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DCIO Debug Displays : DCIO Channel Debug Display

DCIO Channel Debug Display

The DCIO Channel debug display is an interactive display that shows a block diagram of the DCIO pin electronics with the programmable parameters in text boxes. This display lets you view and change the pin electronics, force current/measure voltage (FIMV), and force voltage/measure current (FVMI) values for all DCIO channels.

This section includes the following topics:

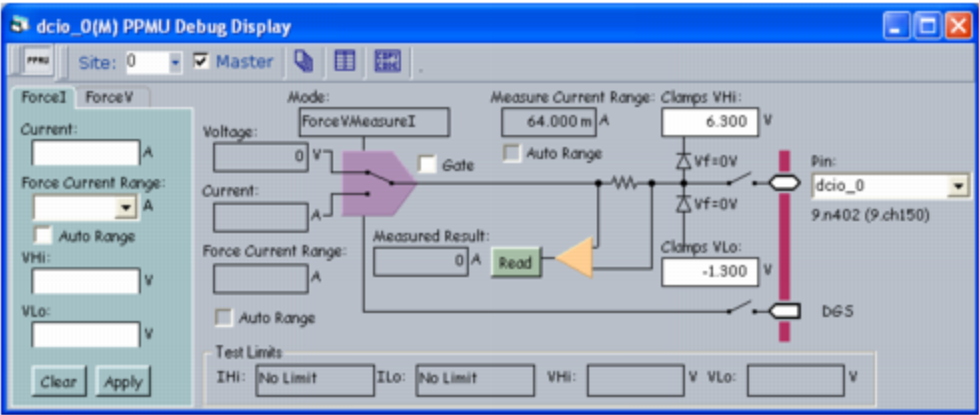
- Starting the DCIO Channel Debug Display
- Starting the DCIO Debug Display After Loading a Test Program
- DCIO Channel Debug Display Interface

Starting the DCIO Channel Debug Display

To bring up the DCIO Channel debug display:

- Load a test program.
- Validate the test program.
- On the IG-XL Tools tab. in the Launch group, select Debug Displays>DCIO Channel.

The DCIO Channel debug display appears and displays the initial values in the drivers (hardware registers). This is shown in the figure below:



DCIO Channel Debug Display

Starting the DCIO Debug Display After Loading a Test Program

1.

Click the Flow Table tab and set a breakpoint.
2.

Run the test program.

The breakpoint automatically causes TDE to start. To manually start TDE, use the Test Debug Environment control in the One-Click Launch group on the IG-XL Tools tab.

3.

In the TDE window, click the TDE menu button and select Instruments>DCIO.

The DCIO Channel Debug Display appears.

Before program execution reaches the breakpoint, the display shows the information it retrieves from the drivers, and all controls are disabled. You cannot change any values until the test program completes or stops at a breakpoint.

DCIO Channel Debug Display Interface

This section describes the toolbar menus and buttons and the fields used in the DCIO Channel debug display. To change values in the debug display, either type the new values in the text box or select the new value from the menu.

The DCIO Channel debug display is used when its pins are in pin electronics mode. This panel displays the following fields and buttons:

DCIO Channel Debug Display Fields

Fields and Buttons	Description
Pin Name	Selects from available DCIO pins.
Mode	Selects the DCIO mode: Pin Electronics, ForceIMeasureV, or ForceVMeasureI.

Vt	Displays or sets the termination input voltage level for the driver.
Vih	Displays or sets the high input voltage level for the driver.
Vil	Displays or sets the low input voltage level for the driver.
Voh	Displays or sets the voltage output threshold for the high comparator.
Vol	Displays or sets the voltage output threshold for the low comparator.
Pin Electronics Mode	Selects pin electronics mode: HiZ or Vt.
Pin Setup	Displays the current state of the DCIO pin and allows you to set the view and to set the pin's initialization state, start state, and stop state. The following are valid values for the pin states: • OFF • HI • LOW • FORCE I • FORCE V • NONE (available for start state only)
Enable Accumulate	Check this box to accumulate a running sum of all measurements in the MRAM. If not checked, all measurements replace previous measured values.
Disable Compare	Sets or clears the Disable expect data compare register. Check sets DCIOPins.DisableCompare = True, and uncheck sets DCIOPins.DisableCompare = False.
Enable Patgen Control	Sets or clears the Channel control using the PatGen register. Check sets DCIOPins.PatternControlEnable = tIOn, and uncheck sets DCIOPins.PatternControlEnable = tIOff.

Pattern Setup

Displays the captured values stored in the measurement RAM (MRAM) and test limits for the selected DCIO pin.

The display shows the values for the following registers, along with the count of the number of measurements taken (the captured values fields are read-only):

Captured Values fields:

- **IHi**: Measure current, expect high
- **IMid**: Measure current, expect mid
- **ILo**: Measure current, expect low
- **VHi**: Measure voltage, expect high
- **VMid**: Measure voltage, expect mid
- **VLo**: Measure voltage, expect low

Test Limits fields:

- **IHi**: Measure current, high limit
- **ILo**: Measure current, low limit
- **VHi**: Measure voltage, high limit
- **VLo**: Measure voltage, low limit

PPMU

Click the PPMU button to open the PPMU debug display for the selected pin. See [PPMU Debug Display](#).



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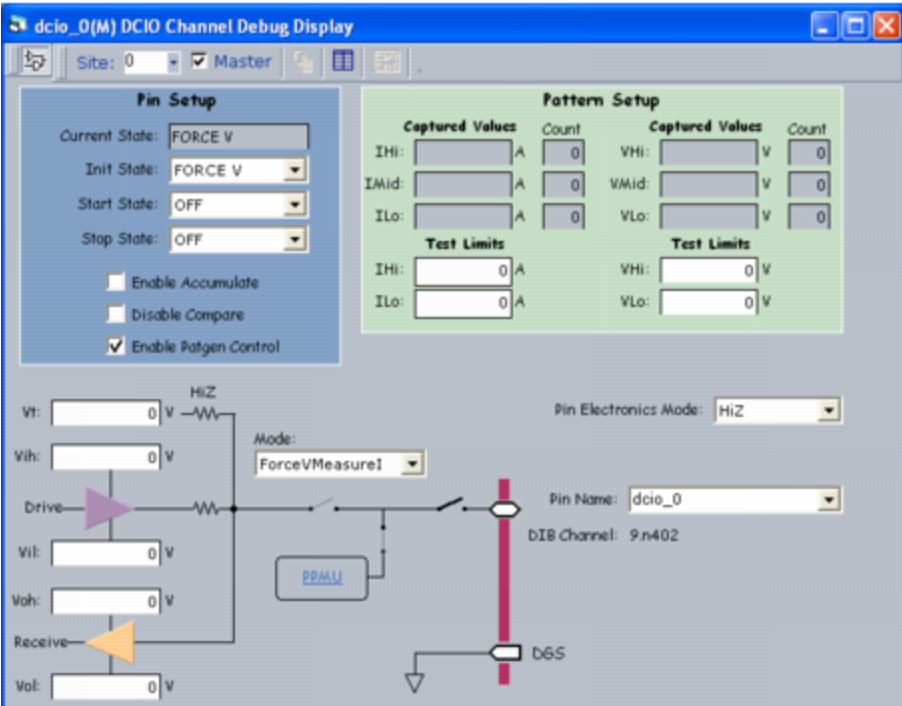
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DCIO Debug Displays : PPMU Debug Display and DCIO Pins

PPMU Debug Display and DCIO Pins

For PPMU programming and debugging, the DCIO instrument uses the same PPMU debug display as the HSD1000. To open the PPMU debug display from the DCIO Channel debug display, click the PPMU debug display button.

Once in the PPMU debug display, you can select both HSD1000 and DCIO pins from the Pin drop-down list in the display. The figure below shows the PPMU display with a DCIO pin.



PPMU Debug Display with DCIO Pin

When debugging a DCIO pin, the selection of current ranges changes to reflect the ranges available on the DCIO. Otherwise, the debug display works the same way as it does when debugging an HSD1000 pin.

☒ **Note:**

The DCIO Channel and PPMU debug displays access the same hardware. As a result, many mode changes to a DCIO pin made in either debug display appear in both displays. For example, if you set a pin to Pin Electronics mode in the DCIO Channel debug display and then apply a force voltage on the pin in the PPMU debug display, you see a mode change to ForceVMeasure1 when you return to the DCIO Channel debug display.

For more information, see [PPMU Debug Display](#).

Using the PPMU Debug Display with Merged PMU Pins

Merged PMU pins use the PPMU Debug Display. You can select a merged PMU pin from the Pin drop-down list in the display. The pin's merge mode appears underneath the pin name.

Force voltage mode is the only valid mode for merged PMU pins, so you can apply force value, current range, and test limits only from the ForceV tab of the debug display. The Measure Current Range drop-down list displays the available current ranges for the specified pin's merge mode. For example, a pin in X4 merge mode displays the following current ranges:

•	32A
•	128A
•	512A
•	2.048mA
•	8.000mA
•	32.000mA
•	256.000mA

For more information, see:

•	Merged PPMU Operation
•	Programming a Merged PPMU Test Instance

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DCIO Programming

DCIO Programming

This section describes programming a DCIO instrument.
The following topics are available:

- [Simulated Configuration Files](#)
- [Programming Procedure Overview](#)
- [DCIO DataTool Programming](#)
- [DCIO Pattern Programming](#)
- [DCIO Test Instance Programming](#)
- [DCIO HRAM Programming](#)
- [DCIO Datalog Setup](#)
- [Using PatternTools with DCIO Patterns](#)
- [Using DCIO in Concurrent Testing](#)

For more information, see:

- [DCIO \(VBT Syntax Reference\)](#)

- [DCIO Examples](#)

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DCIO Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the DCIO offline by forcing IG-XL to map the DCIO. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the DCIO board in slot 15:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]  Type              idprom (type serial rev company)

4.0              HSD-M              979-219-00 0000000 0000-A 0000
                  HSD-M              956-361-00 0000000 0000-A 0000

12.0              HexVS              974-294-00 0000000 0000-A 0000
                  HexVS              956-388-00 0000000 0000-A 0000

15.0              DCIOBoard          974-390-00 0000000 0000-A 0000
```

DCIOBoard

956-448-00 0000000 0000-A 0000

21.0

VSM

974-296-00 0000000 0000-A 0000

VSM

956-361-00 0000000 0000-A 0000

24.0

SupportBoard

974-220-12 0000000 0000-A 0000

SupportBoard

956-354-00 0000000 0000-A 0000

62.0

DistributionBoard

956-355-00 0000000 0000-A 0000

DistributionBoard

956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the DCIO offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the DCIO online, you need not edit any files. IG-XL automatically maps all boards in the tester. For more information, see [Tester Configuration](#).

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DCIO Programming : Programming Procedure Overview

Programming Procedure Overview

You can program the DCIO instrument in several ways. This section covers basic procedures for the following DCIO testing options:

- Computer-Controlled PPMU Testing
- Pattern-Controlled Operation
- Merged PPMU Operation

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DCIO Programming : [Programming Procedure Overview](#) : Computer-Controlled PPMU Testing

Computer-Controlled PPMU Testing

You can use DCIO channels as computer-controlled Pin Parametric Measurement Units (PPMUs). You can also execute PPMU tests through code in the test program flow in the same way as HSD1000 channels.

- | | |
|---|--|
| • | Force Current/Measure Voltage |
| • | Force Voltage/Measure Current |
| • | Seven current ranges (different from HSD1000 PPMU) |
| • | Same programming interface as HSD1000 |

☑ Note:
Do not run a pattern-controlled DCIO test immediately following a computer-controlled PPMU test with the Measurement RAM (MRAM) accumulate setting on. This setup results in IG-XL returning error messages in the output window. This is a known issue that does not have a workaround at this time.

☑ Note:
DCIO does not support computer-controlled PPMU operations while a pattern is running. For computer-controlled PPMU testing, do the following tasks:

- | | |
|----|---|
| 1. | Programming the Pin Map |
|----|---|

2.	Programming the Channel Map
3.	Programming Pin Levels
4.	Programming a PPMU Test Instance

You may also need to do the following:

- [DCIO Datalog Setup](#)

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DCIO Programming : [Programming Procedure Overview](#) : Pattern-Controlled Operation

Pattern-Controlled Operation

You can use patterns to control the DCIO instrument as a digital channel or a PPMU. Note that you can use both of these DCIO modes in a single pattern:

- [Pin Electronics Operation](#)
- [PPMU Operation](#)

Note:

Do not run a pattern-controlled DCIO test immediately following a computer-controlled PPMU test with the Measurement RAM (MRAM) accumulate setting on. This setup results in IG-XL returning error messages in the output window. This is a known issue that does not have a workaround at this time.

For pattern-controlled operation, do the following tasks:

1. [Programming the Pin Map](#)
2. [Programming the Channel Map](#)
3. [Programming Pin Levels](#)
4. [DCIO Pattern Programming](#)
5. [Programming a Functional Test Instance](#)

You may also need to do the following:

- [DCIO HRAM Programming](#)
- [DCIO Datalog Setup](#)
- [Using PatternTools with DCIO Patterns](#)

Pin Electronics Operation

You can use DCIO channels as digital channels that operate with HSD1000 patterns at data rates up to 200kHz. To do this, embed DCIO patterns as instrument microcode inside HSD1000 patterns. The two instruments remain synchronized as the HSD1000 provides the signal that increments the DCIO to its next vector. The DCIO contains 52K of vector memory and 4K of History RAM.

You must program DCIO pin levels just like HSD1000 pins (Vih, Vil, Voh, Vol, and Vt), but no timing is required. DCIO pins can drive high and low, and compare low/high/midband.

PPMU Operation

You can control the DCIO PPMU within a DCIO pattern. Pin data in the pattern instructs individual DCIO pins to:

- [Enter Force Current/Measure Voltage mode](#)
- [Enter Force Voltage/Measure Current mode](#)
- [Return to Pin Electronics mode](#)
- [Make a PMU measurement expecting low, high, or mid](#)

Before the pattern is executed, you must program the force values and test limits in the Pin Levels sheet.

The memory locations available for storing values measured during a pattern are: FVMI Low, FVMI High, FVMI Mid, FIMV Low, FIMV High, and FIMV Mid. You can read these measured values after pattern execution ends.

Instead of storing individual measurements, you can accumulate up to 255 measurements of the same type during a pattern burst. You can use the accumulated value to calculate an average of multiple measurements.

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DCIO Programming : Programming Procedure Overview : Merged PPMU Operation

Merged PPMU Operation

Merged PPMU is a DCIO instrument feature that allows you to merge DCIO channels together to provide higher current capabilities than a single channel. A single DCIO instrument board can simultaneously support single pin and merged channels. When in merged PPMU mode, a DCIO pin can only operate in Force Voltage/Measure Current (FVMI) mode. Merged PPMU channels do not support pin electronics mode or pattern-controlled operation.

For merged PPMU operation, do the following tasks:

- | | |
|----|---|
| 1. | Programming the Pin Map |
| 2. | Programming the Channel Map |
| 3. | Programming Pin Levels (Optional) |
| 4. | Programming a Merged PPMU Test Instance |

You may also need to do the following:

- | | |
|---|---|
| • | Using the PPMU Debug Display with Merged PMU Pins |
|---|---|

For more information, see the following topics:

- | | |
|---|----------------------------------|
| • | DCIO Merged PPMU |
|---|----------------------------------|

- PPMU Channel Merging Rules

PPMU Channel Merging Rules

To use merged PPMU channels, you must follow specific rules for channel assignment. You can merge channels in X2, X4, X8, X16, and X32 configurations. In each merged mode, you can only assign pins to a list of primary channels. Once you assign a primary channel, IG-XL automatically assigns the secondary channels associated with that channel and merge mode to that pin. For example, if you are in merge X4 mode, which merges four channels, you only need to assign channel 0 to a merged PPMU pin because IG-XL automatically assigns channels 1-3.

The following table lists the merge modes, the primary channels, and the maximum current capability of each mode. Note that the primary channel numbers for X32 mode do not increase by 32.

DCIO Merged PPMU Modes

Mode	Primary Channels	Maximum Current
X2	0, 2, 4, ..., 190	128mA
X4	0, 4, 8, ..., 188	256mA
X8	0, 8, 16, ..., 184	512mA
X16	0, 16, 32, ..., 176	1,024mA
X32	0, 48, 96, 144	2,048mA

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DCIO Programming : DCIO DataTool Programming

DCIO DataTool Programming

When using the DCIO instrument for computer-controlled PPMU testing or pattern-controlled operation, you must program a pin map and a channel map. Pin Levels sheet programming is different depending on how you are using the DCIO instrument. This section includes the following topics:

- | | |
|---|---|
| • | Programming the Pin Map |
| • | Programming the Channel Map |
| • | Programming Pin Levels |

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DCIO Programming : DCIO DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. Every test program must have a Pin Map sheet.

Device pins connected to a DCIO instrument can use the following pin types:

I/O	Bidirectional digital pins
Input	Digital input pins, where "input" refers to the DUT pins, so these pins are connected to the input pins of the DUT and send a signal to the DUT.
Output	Digital output pins, where "output" refers to the DUT pins, so these pins are connected to the output pins of the DUT and receive a signal from the DUT.
Power	DCIO merged PPMU pins. For merged PPMU operation, you must add pin names under this pin type into the pin map. Although the DCIO is not a power instrument, IG-XL only allows merging on pins of type Power.

You can create pin groups using DCIO pins, which are collections or sets of pins with some common attribute, such as input pins or output pins. Pin groups can contain other pin groups and/or the following pin types: I/O, Input, and Output. Pin groups with a mix of DCIO and HSD1000 pins are also allowed.

For more information, see [Pin Map Sheet](#) in the DataTool section.

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DCIO Programming : DCIO DataTool Programming : Programming the Channel Map

Programming the Channel Map

A channel map defines how the tester channels (specifically the digital channels) are connected to individual DUT pins. A channel map uses the pin map information and contains the wiring information for the device interface board (DIB). You must include a Channel Map sheet in your test program.

DCIO pins must be assigned to tester channels on a Channel Map sheet, as described below:

Tester Channel Type	List the DCIO pin names and select the DCIO option. For merged PPMU operation, valid tester channel types are: - DCIOMerged2 - DCIOMerged4 - DCIOMerged8 - DCIOMerged16 - DCIOMerged32
Tester Channel Site	Enter the channel number for each pin and site. Channel numbers use the format <slot>.ch<channel>. For example, 6.ch97 refers to channel 97 on the DCIO board in slot 6. Valid channel numbers are from 0 to 191.

For each merged PPMU pin, you only need to specify the primary channel. Enter the name of the pin, and specify the tester channel type based on the merged mode you are using. Once you specify a channel for use as a merged PPMU, the secondary channels associated with that channel and merge mode become unavailable for other pins. Attempting to use them causes a validation error.

For more information, see:

- [Channel Map Sheet](#)
- [Merged PPMU Operation](#)

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DCIO Programming : DCIO DataTool Programming : Programming Pin Levels

Programming Pin Levels

A Pin Levels sheet programs the digital channel levels. The Pin Levels sheet defines values for input and output voltages and current loads to be applied to the DUT.

The DCIO instrument supports incremental levels, which allows you to change levels and driver mode for an individual pin or a pin group that has already been defined on the Pin Levels sheet. IG-XL applies the last set of specified levels for a pin or pin group when adding a new pin or pin group, applying and sequencing levels, characterizing pins, and determining specification dependencies. With this flexible programming model, you can reassign new pin levels to existing pins on the sheet while preserving previous pin level setups. You can always revert to previous levels settings and behavior by removing the most recent pin level change on the sheet.

For the DCIO instrument, you must assign a pin levels parameter for each pin name or pin group. The following table lists the DCIO pin level Parameter and Value fields. If a pin does not have levels specified, a validation error occurs.

Pin Levels sheet programming is optional for merged PPMU pins.

Parameter	An entry is required for each pin and pin group. DCIO pins have two groups of pin levels. The first group contains levels that DCIO pins use in Pin Electronics mode: • Vil: Forced low input voltage level for the driver • Vih: Forced high input voltage level for the driver • Vol: Threshold voltage output reference for the low comparator
-----------	---

- **Voh**: Threshold voltage output reference for the high comparator
- **Vt**: Forced termination input voltage level for the driver

The second group contains levels that DCIO pins use in pattern-controlled PPMU mode:

- **ForceV**: Force voltage value for pattern-controlled PPMU
- **Forcel**: Force current value for pattern-controlled PPMU
- **ComparelHi**: Output high threshold for pattern-controlled current measurements
- **ComparelLo**: Output low threshold for pattern-controlled current measurements
- **CompareVHi**: Output high threshold for pattern-controlled voltage measurements
- **CompareVLo**: Output low threshold for pattern-controlled voltage measurements
- **ForcelRange**: Current range for pattern-controlled force current/measure voltage operation
- **MeasurelRange**: Current range for pattern-controlled force voltage/measure current operation
- **ClampVHi**: High voltage clamp level
- **ClampVLo**: Low voltage clamp level
- **DriverMode**: Driver termination mode

Merged PPMU operation, which is optional for this sheet, only supports Force Voltage/Measure Current (FVMI) mode, so only the following values are available:

- ForceV
- ComparelHi
- ComparelLo
- MeasurelRange

Value

Enter the value for the parameter as either a numeric value or a formula. If you enter a number, you can use a formula to include engineering units: for example, $=200\text{mV}$.

For ForcelRange, the values are:

- .032 (32mA)
- .008 (8mA)
- .002 (2mA)
- .000512 (512A)
- .000128 (128A)
- .000032 (32A)
- .000008 (8A)

For MeasurelRange, the values are:

- .064 (64mA)

- [.008 \(8mA\)](#)
- [.002 \(2mA\)](#)
- [.000512 \(512A\)](#)
- [.000128 \(128A\)](#)
- [.000032 \(32A\)](#)
- [.000008 \(8A\)](#)

For [DriverMode](#), the values are:

- [HiZ](#): Unterminated
- [VT](#): Terminate to VT through 1K resistor

Default values are inserted for the parameters that have global spec symbols automatically created on the [Global Specs](#) sheet.

For more information, see:

- | |
|------------------------------------|
| • Pin Levels Sheet |
|------------------------------------|

- | |
|---|
| • Incremental Levels Mode |
|---|

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DCIO Programming : DCIO Pattern Programming

DCIO Pattern Programming

Patterns contain the data that DCIO channels send to the DUT. Patterns also contain instructions, called microcodes, that determine test program flow. All patterns in a burst must contain the same list of instruments.

This section describes how to create DCIO test patterns in an ASCII text file using the digital pattern and DCIO-specific microcodes. This section includes the following topics:

- | | |
|---|---|
| • | Creating a DCIO Pattern File |
| • | Synchronizing DCIO and HSD1000 Patterns |

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DCIO Programming : DCIO Pattern Programming : Creating a DCIO Pattern File

Creating a DCIO Pattern File

To create a DCIO pattern file, you must embed DCIO patterns as instrument microcode inside HSD1000 patterns. The DCIO instrument does not have its own pattern generator but uses the HSD1000 pattern generator to increment through its vectors. For more information, see:

- HSD1000 Test Pattern Programming.
- About Pattern Language.

Pattern Header Requirements

You must declare the DCIO pins in the instruments declaration of the pattern header. Declare the **instrument type** as `dcio`. Place the instruments statement after the import and export statements and before the `vm_vector` statement. In the following example, the pattern uses four pins named `DCIO_0`, `DCIO_1`, `DCIO_2`, and `DCIO_3`.
`instruments = {(DCIO_0, DCIO_1, DCIO_2, DCIO_3): dcio;}`

For more information, see [Instruments Statement](#) in the Pattern Language section.

DCIO Vectors and Microcode

A DCIO vector can appear on any HSD1000 pattern vector. The microcode for each DCIO vector contains data for each DCIO pin. The following pin states are valid:

DCIO Pin State Microcode	
Pin State Microcode	Description
0	Drive Vil/Enter PE Mode

1	Drive Vih/Enter PE Mode
M	Expect Midband
L	Expect Low
H	Expect High
X	Drive VT/Drive HiZ/Enter PE Mode
C	Enter FIMV Mode
K	Enter FVMI Mode

Note that some pin state microcodes perform different functions depending on the DCIO operation mode. Opcodes 0, 1, X, C, and K can change the pin mode. The following table lists the function of each pin state microcode in each DCIO mode:

Pin State Microcode and DCIO Pin Modes

Pin State	PE Mode	FIMV Mode	FVMI Mode
0	Drive 0	Enter PE Mode, Drive Vil	Enter PE Mode, Drive Vil
1	Drive 1	Enter PE Mode, Drive Vih	Enter PE Mode, Drive Vih
M	Expect Midband	Measure, Expect Midband	Measure, Expect Midband
L	Expect Low	Measure, Expect Low	Measure, Expect Low
H	Expect High	Measure, Expect High	Measure, Expect High
X	Drive VT	Enter PE Mode	Enter PE Mode
C	Enter FIMV Mode	Do Nothing	Enter FIMV Mode
K	Enter FVMI Mode	Enter FVMI Mode	Do Nothing

The code sample below shows part of a DCIO pattern with four DCIO pins and six HSD1000 pins:

```
digital_inst = hsdm;
opcode_mode = single;
import tset tset_1;
instruments = {(dcio_0, dcio_1, dcio_2, dcio_3): dcio;}
vm_vector DCIO_patt( $tset, HSD_0, HSD_1, HSD_2, HSD_3, HSD_4, HSD_5 )
{
> tset_1 0 0 0 0 0 0;
> tset_1 0 0 0 0 0 0;
(( dcio_0, dcio_1, dcio_2, dcio_3):dcio = 1M1M)
```

```
> tset_1 0 0 0 0 0 0;  
repeat 500 > tset_1 0 0 0 0 0 0;  
(( dcio_0, dcio_1, dcio_2, dcio_3):dcio = 0L0L)  
    > tset_1 0 0 0 0 0 0;  
repeat 50 > tset_1 0 0 0 0 0 0;
```

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DCIO Programming : DCIO Pattern Programming : Synchronizing DCIO and HSD1000 Patterns

Synchronizing DCIO and HSD1000 Patterns

An HSD1000 pattern increments a DCIO pattern to the next vector whenever a DCIO microcode instruction appears in the pattern. However, since the DCIO instrument can run at a maximum of 200kHz, the HSD1000 pattern must loop and wait to give the DCIO enough time to complete its vector before continuing execution. For the required minimum wait times when changing pin mode (switching between PE, FIMV, and FVMI), see the table in [DCIO Cycle Periods](#). The minimum wait time for all other operations (drive, strobe, and measure) is 5s.

After each DCIO vector, the HSD1000 must repeat the proper number of vectors to meet the minimum wait times as defined above. In the following example, the HSD1000 cycle period is 100ns, so you must include a repeat of at least 500 after a DCIO mode change:

```
(( dcio_0, dcio_1, dcio_2, dcio_3):dcio = CCCC) // Enter FIMV Mode
> tset_1 0 0 0 0 0 0;
repeat 500 > tset_1 0 0 0 0 0 0;           // Mode change requires 50us
(( dcio_0, dcio_1, dcio_2, dcio_3):dcio = MMMM) // Measure voltage
> tset_1 0 0 0 0 0 0;
repeat 50 > tset_1 0 0 0 0 0 0;           // Measure requires 5us
```

The HSD1000 halt microcode also halts the DCIO pattern. You must make sure that at least 5s elapse between the final DCIO vector and the halt opcode to ensure that the DCIO vector completes execution.

	
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DCIO Programming : DCIO Test Instance Programming

DCIO Test Instance Programming

You can use the instance editor to create and edit Functional_T and PinPMU_T test instances. A test instance defines a test that is applied to the DUT when the test program runs. You can create as many test instances as you want. The Test Instances sheet lists all test instances for a test program. To create or edit a Functional_T or PinPMU_T test instance in the instance editor, see [Creating and Editing Test Instances](#). Once you have created a test instance, program the Functional_T or PinPMU_T test instance by filling in the parameter fields. For a reference to Functional test parameters, see [Functional VBT Test](#). For PPMU test parameters, see [Pin PMU VBT Test](#). Depending on how you are using the DCIO instrument, you must program one of the following test instance types:

- | | |
|---|---|
| • | Programming a PPMU Test Instance |
| • | Programming a Functional Test Instance |
| • | Programming a Merged PPMU Test Instance |

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DCIO Programming : DCIO Test Instance Programming : Programming a PPMU Test Instance

Programming a PPMU Test Instance

To use DCIO channels as computer-controlled PPMUs, you must program a PPMU test instance. For a reference to PPMU template parameters found in the Pin PMU instance editor, see [Pin PMU VBT Test](#) in the VBT Test Library section.

Programming a computer-controlled PPMU test instance using the DCIO instrument is nearly identical to programming a PPMU test instance for the HSD1000 instrument. The HSD1000 PPMU test instance is used for DCIO testing. DCIO-specific programming occurs in the following places:

Pins	You can specify DCIO pins and pingroups, as well as a mix of DCIO and HSD1000 pins. Optionally, you can specify DCIO pins in the DriveLoPins, DriveHiPins, DriveZPins, and FloatPins fields in the Pin States section. See Pin States Parameters.
ForcelRange MeaslRange	The drop-down lists ranges for all available instruments in the test system. When selecting a range for one instrument, IG-XL automatically selects the nearest range for the other instrument. For example, if you test both DCIO and HSD1000 pins and choose the 512uA current range, IG-XL sets the DCIO pins to 512uA and the HSD1000 pins to the 2mA range.

When performing a computer-controlled PPMU test instance, note that while programming of the DCIO and HSD1000 may be the same, the instruments do have hardware differences. One significant difference is the voltage versus current characteristics between the two instruments. These instruments have different current sourcing and sinking capabilities at different voltages.

		
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DCIO Programming : [DCIO Test Instance Programming](#) : Programming a Functional Test Instance

Programming a Functional Test Instance

To use DCIO channels as digital channels in pattern-controlled operation, you must program a functional test instance. For a reference to functional template parameters found in the Functional instance editor, see [Functional VBT Test](#) in the VBT Test Library section.

No special procedures are required for creating a functional test instance. To create a DCIO functional test, you must include the following in a standard functional test instance:

- [A pattern with DCIO vectors](#)
- [A level with DCIO pin levels specified](#)

Optionally, you can specify DCIO pins in the [DriveLoPins](#), [DriveHiPins](#), [DriveZPins](#), and [FloatPins](#) fields in the Pin States section. For more information, see [Pin States Parameters](#).

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DCIO Programming : [DCIO Test Instance Programming](#) : Programming a Merged PPMU Test Instance

Programming a Merged PPMU Test Instance

To create a PPMU test instance with merged PPMU pins, use a standard Pin PMU test template. For a reference to PPMU template parameters found in the Pin PMU instance editor, see [Pin PMU VBT Test](#) in the VBT Test Library section.

To address specific merged PPMU pin issues, you must enter the correct information in the following template fields:

Pins	Enter a merged PPMU pin name or pin group in this field to create a Pin PMU test.
MeasureMode	When using merged PPMU pins, you must set this field to Current , since this is the only operation permitted.
ForceRange MeasRange	If the Pins contains a merged PPMU pin, the list of available current ranges expands to include the higher current capabilities supported by the merged PMU pins.

Note:

For best performance, always use the highest available current range for the merged mode you are using.

For more information, see:

- [Merged PPMU Operation](#)
- [Using the PPMU Debug Display with Merged PMU Pins](#)

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DCIO Programming : DCIO HRAM Programming

DCIO HRAM Programming

The DCIO instrument provides 4K of HRAM behind each DCIO channel. DCIO HRAM setup is similar to the HSD1000 and uses most of the same setup parameters. Note that HRAM sizes differ between the DCIO and HSD1000.

The DCIO HRAM setup follows the same procedure as HSD1000 HRAM. Use one of the following methods to set up the DCIO HRAM for capture:

- Setting Up DCIO HRAM Using DataCollect Setup

- Setting up DCIO HRAM Using VBT

In DCIO HRAM setup, you can specify the following:

- Trigger on first cycle or on first failing cycle

- Wait for PatGen event before triggering. PatGen events include vector location, cycle number, and pattern/module repeat count.

- Capture all vectors, failing vectors, passing vectors, vectors with STV, first failing vector in a module, first passing vector in a module, or capture nothing

- Number of pretrigger cycles to display

 **Note:**

If you set the HSD1000 HRAM halt mode to stop on HRAM full or on first fail, these two conditions can stop a DCIO vector that has not completed execution.

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[DCIO Programming](#) : [DCIO HRAM Programming](#) : Setting Up DCIO HRAM Using DataCollect Setup

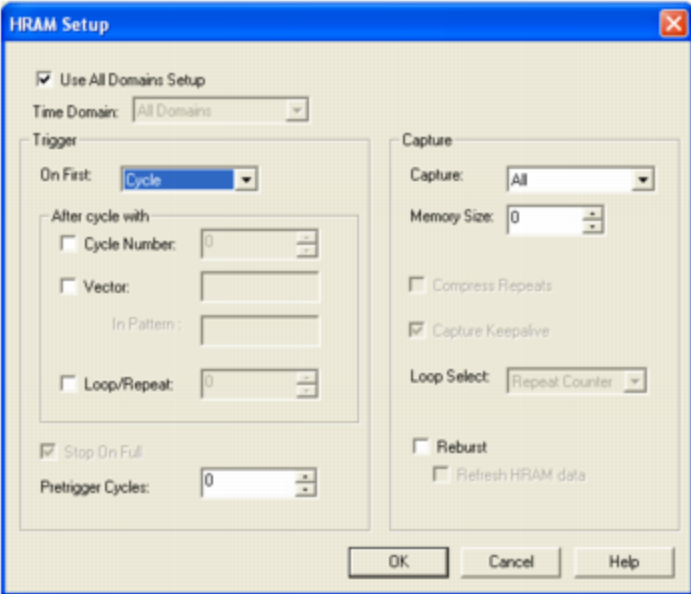
Setting Up DCIO HRAM Using DataCollect Setup

Inside IG-XL, you can set up HRAM by creating a custom datalog setup. To do this, follow these steps:

1. [Open the DataCollect Setup window by clicking](#) DataCollect Setup in the Data Collect group on the IG-XL Main tab.
2. [Check the](#) Customize box in the Setups tab of the panel.
3. [Click the](#) Setup File button to open the custom datalog setup panel.

[The bottom of this panel contains the HRAM section.](#)

4. [Click the](#) Full Edit button to access all HRAM setup options. The figure below shows this window.



Custom Datalog Setup Window for HRAM

Creating an HRAM setup in this way programs both the HSD1000 HRAM and the DCIO HRAM at the same time. This means that you must use care in setting up PatGen conditions because any settings you mean to apply to one instrument’s HRAM will also be applied to the other instrument. For more information, see HRAM Setup in DataCollect Setup.

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DCIO Programming : DCIO HRAM Programming : Setting up DCIO HRAM Using VBT

Setting up DCIO HRAM Using VBT

You can use VBT code to create functions that set up the DCIO HRAM. The programming syntax of the DCIO HRAM is similar to the HSD1000 HRAM. You can access it through the following VBT node:

TheHdw.DCIO.HRAM

All properties and methods for this node work the same as the HSD1000 HRAM, except for Size, which has a higher maximum value due to the DCIO's larger HRAM depth, and LookupHSDVector, which returns the offset of HSD1000 vector that initiated the DCIO vector.

For more information, see [HRAM](#) in the VBT Syntax Reference.

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DCIO Programming : DCIO Datalog Setup

DCIO Datalog Setup

You can send DCIO test results to the standard datalog output streams, including the datalog output window, STDF files, and text files. The same data collection setup applies to both HSD1000 and DCIO pins.

- | | |
|---|--|
| • | DCIO Datalogs for Computer-Controlled PPMU Tests |
| • | DCIO Datalogs for Functional Tests |
| • | DCIO Datalogs for Pattern-Controlled PPMU Tests |
| • | Reading DCIO PPMU Measurement RAM Registers |

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DCIO Programming : [DCIO Datalog Setup](#) : DCIO Datalogs for Computer-Controlled PPMU Tests

DCIO Datalogs for Computer-Controlled PPMU Tests

Computer-controlled PPMU tests display DCIO pin output that is identical to HSD1000 pins. The following is a simple example:

Number	Site	Test Name	Measured
108	0	DCIO_MV	-698.1925 mV
109	0	DCIO_MV	-1.0713 V
110	0	DCIO_MV	-666.5944 mV (F)
111	0	DCIO_MV	-601.7877 mV (F)

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DCIO Programming : DCIO Datalog Setup : DCIO Datalogs for Functional Tests

DCIO Datalogs for Functional Tests

In short datalog format, functional tests report a combined pass or fail result for a DCIO pattern. If any HSD1000 or DCIO pins fail during a pattern burst, IG-XL reports the test as a fail.

Expanded datalog output during a functional test can display DCIO vectors interleaved with HSD1000 vectors. By default, the expanded functional datalog displays failing vectors that have been stored in HRAM of both instruments. Depending on the HRAM settings, a vector with both DCIO and HSD1000 data can appear in the datalog. In these cases, the DCIO and HSD1000 HRAM data appear on separate lines in the datalog.

The following datalog sample shows three failing vectors from a pattern containing eight DCIO pins and six HSD1000 pins. The first vector in this sample shows fails only in the HSD1000 HRAM. The second vector shows fails only in the DCIO HRAM. The final vector appears twice in the datalog: the first time for the HSD1000 HRAM and the second time for the DCIO HRAM.

Number	Site	Test Name	Pattern	1st Failed Cycle	Total Failed Cycles
<PE_FI_IOSwitch>					
105	0	PE_FI_DCIO	DCIO_Patt (F)	0	19

PINS

dddddddddhhhhh
ccccccccsssss
iiiiiiiiddddd
oooooooo_____
_____012345
01234567

Cycle Relative Repeat

898 386 0 *****HHHHHH
-> (F) *****LLLLLL

N/A 390 0 HHHHHHHH*****
-> (F) HGHHHLHH*****

928 392 0 *****HHHHHH
-> (F) *****LLLLLL

N/A 392 0 HHHHHHHH*****
-> (F) HHHHHLHH*****

IG-XL aligns DCIO and HSD1000 vectors based on the vector time stamp, which is taken at the time of the strobe. Since the DCIO strobe occurs approximately 4.8s after the start of the DCIO vector, the datalog might display many HSD1000 vectors before the DCIO vector appears. Due to the 16ns accuracy of the DCIO time stamper, DCIO vectors might not exactly line up with HSD1000 vectors. The exact location of the DCIO vector in a datalog can vary from run to run.

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DCIO Programming : DCIO Datalog Setup : DCIO Datalogs for Pattern-Controlled PPMU Tests

DCIO Datalogs for Pattern-Controlled PPMU Tests

If you make pattern-controlled PPMU measurements during a functional pattern burst, you must call an interpose function after the test that sends the PPMU measurements to the datalog. Otherwise, only the pass/fail results of the PPMU measurements appear in the functional portion of the datalog output. Datalog the PPMU measurements using the following VBT call:

```
TheHdw.DCIO.Pins(PinList).MRAM.test(Type)
```

You can specify any of the six MRAM registers by the Type parameter, or you can display all six. Only the pins and registers for which measurements have been made appear in the datalog. If you use the PostPatF interpose point, IG-XL displays the MRAM results before the functional portion of the datalog. If you use the EndOfBodyF interpose point, IG-XL displays the MRAM results after the functional datalog.

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DCIO Programming : DCIO Datalog Setup : Reading DCIO PPMU Measurement RAM Registers

Reading DCIO PPMU Measurement RAM Registers

When you make a DCIO PPMU measurement from within a pattern, IG-XL stores the measured value in one of six available measurement RAM (MRAM) registers. The DCIO has one MRAM register for each combination of measure mode and strobe type:

- Measure Current, Expect High
- Measure Current, Expect Low
- Measure Current, Expect Mid
- Measure Voltage, Expect High
- Measure Voltage, Expect Low
- Measure Voltage, Expect Mid

At the start of each pattern burst, IG-XL clears these registers in preparation for new measurements. You can either send the measured values to the datalog, or read them into user-defined variables. The following syntax provides access to the MRAM registers:

```
TheHdw.DCIO.Pins(PinList).MRAM.Read(Type, Value, Count, Result)
```

Since this command returns data as PinListData objects, you generally must execute this code inside site and pin loops to read back measurements for individual channels. The following code example reads three voltage measurements from the pin "dcio_pin":

[Dim HighVal As New PinListData](#)

[Dim LowVal As New PinListData](#)

[Dim MidVal As New PinListData](#)

[TheHdw.DCIO.Pins\("dcio_pin"\).MRAM.Read tIDCIOMramVoltageHigh, HighVal](#)

[TheHdw.DCIO.Pins\("dcio_pin"\).MRAM.Read tIDCIOMramVoltageLow, LowVal](#)

[TheHdw.DCIO.Pins\("dcio_pin"\).MRAM.Read tIDCIOMramVoltageMid, MidVal](#)

[For more information, see TheHdw.DCIO.Pins.MRAM.](#)

[DCIO MRAM Accumulate Mode](#)

You can use the MRAM to store data with the accumulate feature turned on or off. When accumulate is off, each register only stores the value of the last measurement made. IG-XL overwrites any previous measurements. When accumulate is on, IG-XL adds measured values to the existing value in the register so that the register stores a running sum of all measurements. The register also records the number of measurements so that IG-XL can calculate an average value. The following syntax enables the accumulate mode:

[TheHdw.DCIO.Pins\(PinList\).MRAM.Accumulate = tOn | tOff](#)

Note:

Do not run a pattern-controlled DCIO test immediately following a computer-controlled PPMU test with the Measurement RAM (MRAM) accumulate setting on. This setup results in IG-XL generating error messages in the output window. This is a known issue that does not have a workaround at this time.

		
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DCIO Programming : Using PatternTools with DCIO Patterns

Using PatternTools with DCIO Patterns

You can use both the standard PatternTool and the HRAM PatternTool with DCIO patterns. For more information, see [About PatternTool](#).

Viewing and Editing DCIO Patterns in PatternTool

PatternTool can be used for both HSD1000 and DCIO patterns. DCIO vectors appear together with the HSD1000 vectors on which they are located. DCIO pins appear in the leftmost pin columns in the PatternTool vertical display. For vectors that have no DCIO data, the DCIO pin columns will be blank. DCIO data can be edited, added or deleted through PatternTool.

DCIO_Module_FIEven_MIOdd.PAT - DCIO_FIEven_MIOdd															
				-											
				D	D	D	D	D	D	D	D				
				C	C	C	C	C	C	C	C				
				I	I	I	I	I	I	I	I				
				O	O	O	O	O	O	O	O				
				0	1	2	3	4	5	6	7	H	H	H	H
				(	(	(	(	(	(	(	(	S	S	S	S
				D	D	D	D	D	D	D	D	D	D	D	D
TSet	Command	Label	Vector/ Cycle	C	C	C	C	C	C	C	C	0	1	2	3
tset_1			0									0	0	0	0
tset_1			1									0	0	0	0
tset_1			2									0	0	0	0
tset_1			3									0	0	0	0
tset_1			4									0	0	0	0
tset_1			5									0	0	0	0
tset_1			6	C	K	C	K	C	K	C	K	0	0	0	0
tset_1	repeat 500		7									0	0	0	0
tset_1			8	C	L	C	M	C	H	C	L	0	0	0	0
tset_1	repeat 25		9									0	0	0	0

DCIO and HSD1000 Patterns in PatternTool

PatternTool supports all operations for DCIO except for actual mode. For more information, see [Opening and Managing Pattern Files](#).

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DCIO Programming : Using DCIO in Concurrent Testing

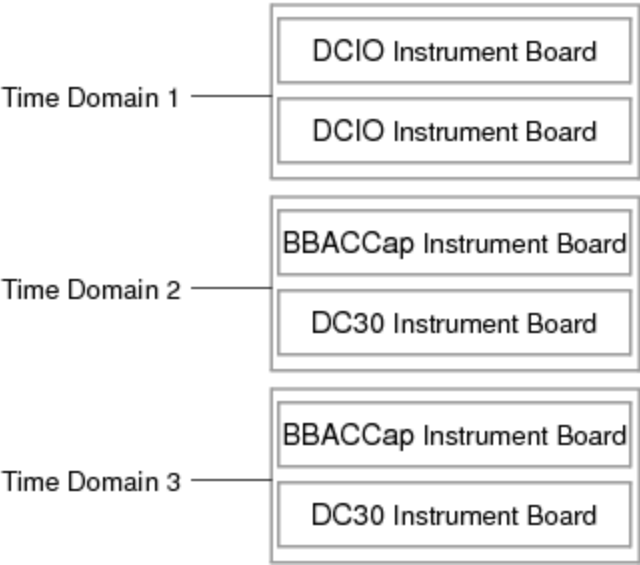
Using DCIO in Concurrent Testing

To support concurrent testing, you can assign DCIO channels to a time domain. This means that DCIO instruments can be part of a concurrent pattern burst. However, only one pattern in the concurrent burst can contain DCIO instruments in it at one time. This limitation means that DCIO channels cannot be in more that one time domain for each concurrent burst.

Concurrent features for DCIO include:

- A DCIO board can be configured for one time domain
- Multiple DCIO instruments can be grouped into a common time domain
- DCIO is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with two DCIO instruments assigned to its one time domain and other UltraFLEX instruments assigned to other time domains.



DCIO Concurrent Testing Configuration Example
For more information, see [About Concurrent Test](#).

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DCIO Theory of Operation

DCIO Theory of Operation

This section provides a functional description of how the DCIO works, including information on the following topics:

- [DCIO Hardware](#)
- [DCIO Pin Electronics and PPMU](#)
- [DCIO Timing](#)
- [DCIO Pattern Architecture](#)
- [DCIO History RAM \(HRAM\)](#)
- [DCIO Merged PPMU](#)

See also:

- [About the DCIO](#)
- [DCIO Programming](#)
- [DCIO Debug Displays](#)

•	DCIO Examples
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For information on DCIO instrument performance specifications and standards, see [DCIO Specifications](#).

For DCIO DIB design guidelines, see [DCIO DIB Design Guidelines](#).

		
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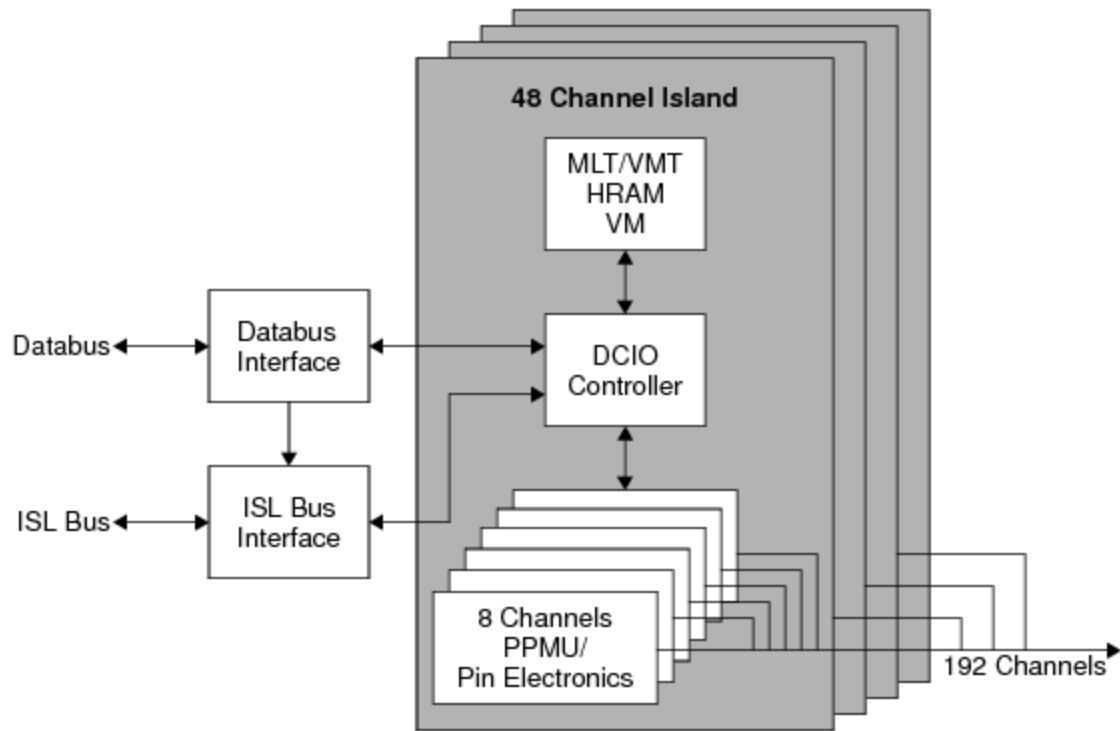
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DCIO Theory of Operation : DCIO Hardware

DCIO Hardware

The following is a high-level block diagram of the DCIO board and its major functions. Each DCIO board contains four 48 channel "islands." Each island contains the following:

- DCIO controller, which manages all DCIO operations
- Memory, which consists of the module lookup table (MLT), vector module table (VMT), History RAM (HRAM), and the vector memory (VM)
- PPMU and pin electronics: Each island supports six modules of eight DCIO channels for 48 channels per island and 192 channels per DCIO board.



DCIO Hardware Block Diagram

The following topics are available:

- [DCIO Pin Electronics and PPMU](#)

- [DCIO Timing](#)

- [DCIO Pattern Architecture](#)

- [DCIO History RAM \(HRAM\)](#)

- [DCIO Merged PPMU](#)



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DCIO Theory of Operation : DCIO Pin Electronics and PPMU

DCIO Pin Electronics and PPMU

DCIO channels can operate as 200kHz pin electronics and as a per pin PMU (PPMU). The DCIO channel operates in different modes to support different operation applications.

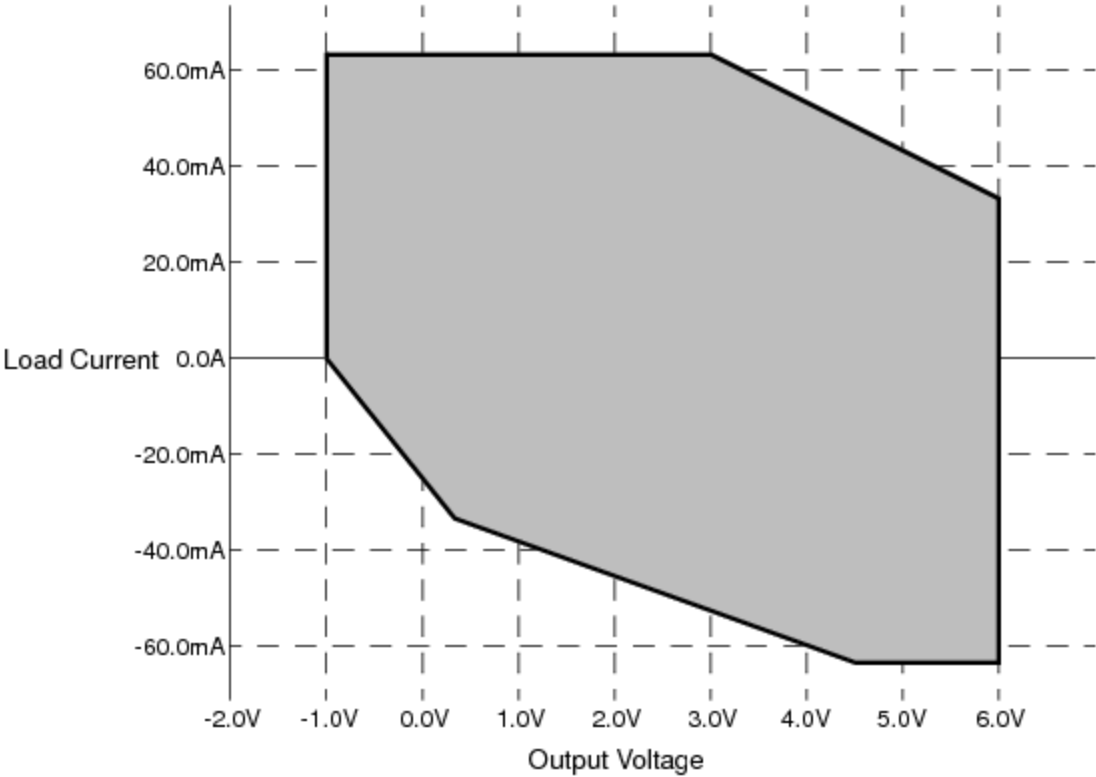
DCIO Operation Modes	
Mode	Applications
Force Voltage/Measure Voltage	Pin electronics mode
Force Voltage/Measure Current (FVMI)	Computer-controlled PPMU, Pattern-controlled PPMU, Merged PPMU
Force Current/Measure Voltage (FIMV)	Computer-controlled PPMU, Pattern-controlled PPMU

The DCIO channel operates as a PPMU with the following current ranges:

- 8A
- 32A
- 128A
- 512A

- 2mA
- 8mA
- 32mA (Force current only)
- 64mA (Measure current only)

The following figure shows the V/I curve at 64mA for the DCIO used as a PPMU.



V/I Curve for DCIO PPMU, 64mA Range (not to scale)

DCIO Drivers

The DCIO can operate as a PPMU in both Force Voltage/Measure Current (FVMI) and Force Current/Measure Voltage (FIMV) modes. In pin electronics mode, the DCIO drivers can drive V_{ih} or V_{il} on a per cycle basis. In addition, the drivers can go to HiZ or drive V_t through a 1K (typical) termination resistor. The following specifications apply to the DCIO channel in pin electronics mode and in PPMU FVMI Mode

Voltage Range	-1V to +6V
(V_{il} , V_{ih} , PPMU FV)	
Voltage Range (V_t)	-1V to +6V

Resolution	1mV
Accuracy	10mV

The following specifications apply when in FIMV mode:

Measure Voltage Range	-1V to +6V
Resolution	0.6mV
Accuracy	0.2% + 14mV

DCIO Comparators

In pin electronics mode, the DCIO comparators can compare at fixed low, high, and midband levels (valid is not supported). A single edge strobe is available in each DCIO cycle. The following specifications apply to the comparators:

Vol/Voh Range	-1V to +6V
Resolution	1mV
Accuracy (0 to 1.2V)	10mV
Accuracy (-1 to 6V)	0.2% + 10mV

DCIO Over-current Protection

DCIO channels provide over-current protection, which clamps the current at a fixed maximum of 75mA. You can enable or disable the protection on each pin individually.

DCIO Voltage Clamps

The DCIO provides voltage clamps for use in Force Current/Measure Voltage (FIMV) PPMU mode.

PPMU Vclamp Low Range	-1.5V to 0V
PPMU Vclamp High Range	0V to +6.3V
PPMU Clamp Level Accuracy	1.0% + 300mV
Resolution	1mV

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DCIO Theory of Operation : DCIO Timing

DCIO Timing

The DCIO timing system involves the following elements:

- DCIO Cycle Periods
- DCIO Drive and Compare Timing

For more information, see HSD1000 Timing.

DCIO Cycle Periods

When you use DCIO channels as digital pin electronics, the timing is fixed and not programmable. The DCIO does not contain an independent pattern generator and instead gets the signal to increment to the next vector from an HSD1000 board through the ISL bus. You must ensure that increments from the HSD1000 board do not arrive at a faster rate than the DCIO board can sustain. The minimum period for completion of a DCIO cycles is 5s.

You can use the DCIO as a purely DC instrument, setting pins to desired levels and leaving them in that state for the duration of a test.

Each DCIO cycle requires a minimum of 5s to complete. Switching between the different operating modes of the DCIO channel requires more time. The mode switching times range from 60s to over 1ms and vary depending on what current range is being used. The smaller current ranges take longer to switch.

The table below lists the minimum period required for a cycle with a mode change.

Note:

Actual switching times depend upon the characteristics of the DUT. While the numbers in the table below show the minimum time required by the test system, the device can require additional time to settle after a mode change.

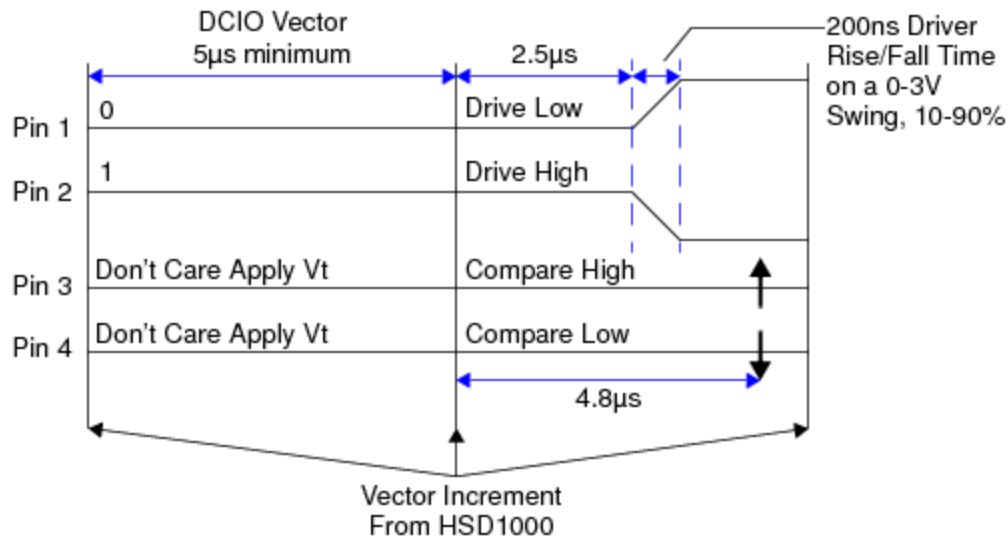
Required Minimum Period for Cycles with Mode Changes

Mode Change	Current Range	Minimum Period
PE to PMU FVMI	64mA	80s
	8mA	80s
	2mA	80s
	512A	80s
	128A	180s
	32A	375s
	8A	850s
PE to PMU FIMV	32mA	190s
	8mA	160s
	2mA	160s
	512A	180s
	128A	430s
	32A	1615s
	8A	6280s
FVMI PMU to PE	32mA	80s
	8mA	80s
	2mA	80s
	512A	80s
	128A	80s
	32A	80s
	8A	80s
FIMV PMU to PE	32mA	80s
	8mA	80s
	2mA	80s
	512A	80s
	128A	80s

	32A	80s
	8A	80s
FVMI PMU to FIMV PMU	32mA	200s
	8mA	200s
	2mA	200s
	512A	200s
	128A	450s
	32A	1600s
	8A	6250s
FIMV PMU to FVMI PMU	64mA	80s
	8mA	80s
	2mA	80s
	512A	80s
	128A	200s
	32A	400s
	8A	825s

DCIO Drive and Compare Timing

During a DCIO drive or force cycle, a fixed amount of time is required for the channel to reach the programmed level. When not switching modes, the channel begins to drive the programmed state after a 2.5s latency. The rise/fall time of a DCIO channel is 200ns for a 3V programmed swing. Each DCIO pin has one strobe per cycle. The strobe timing is fixed at 4.8s from the start of the cycle and cannot be changed. The following figure shows the DCIO drive and compare timing.



DCIO Drive and Compare Timing

Same cycle I/O is not supported. Therefore, in each cycle, a DCIO channel can drive or compare, but not both.

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DCIO Theory of Operation : DCIO Pattern Architecture

DCIO Pattern Architecture

Each DCIO board contains 54K locations of vector memory (VM). Each location in vector memory stores pattern pin data. You can set ignore, store, and care bits on a per-vector basis and three bits of pin data for each pin. Looping or branching is not allowed in DCIO vectors. DCIO has no minimum pattern size requirement.

The DCIO board does not contain any subroutine memory (SRM).

The DCIO also contains a vector module sequencer (VMS) that manages the pattern executions. The VMS consists of two tables, the module lookup table (MLT) and the vector module table (VMT). The MLT stores the start address and size of each pattern module. The VMT controls the order of pattern module execution. The DCIO contains 128K of VMT and 54K of MLT.

For more information, see:

- HSD1000 Pattern Architecture
- DCIO Pattern Programming

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DCIO Theory of Operation : DCIO History RAM (HRAM)

DCIO History RAM (HRAM)

The History RAM (HRAM) hardware is used to capture and provide vector fail information. The datalogging and debugging tools can then use the captured data for analysis. The DCIO has 4K locations of HRAM that is independent of the HSD1000 HRAM.

The DCIO HRAM captures the following data:

- | | |
|---|--------------------------|
| • | DCIO pin data |
| • | DCIO vector number |
| • | Module count |
| • | Module repeat count |
| • | Pattern set repeat count |

The DCIO HRAM triggering and capturing mechanism works the same way as on the HSD1000 HRAM. However, the two instruments’ trigger and capture setups can be different because the DCIO hardware is independent from the HSD1000.

Because the DCIO pattern stop signal comes from an HSD1000 board, DCIO patterns cannot stop due to the DCIO HRAM becoming full.

For more information, see:

- | | |
|---|----------------------------|
| • | HSD1000 History RAM (HRAM) |
|---|----------------------------|

•	DCIO HRAM Programming
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DCIO Theory of Operation : DCIO Merged PPMU

DCIO Merged PPMU

You can merge up to 32 DCIO channels together to create a high current PPMU. With the maximum merging, a DCIO pin can drive up to 2.048A. The current capability of a merged PPMU pin is the number of merged channels multiplied by 64mA. Therefore, the maximum current capability of a X32 merged PMU pin is 32 X 6mA, or 2.048A. You can set the current range in any of the following places:

- Pin Levels sheet
- Pin PMU Template
- PPMU Debug Display
- VBT code

The following PPMU merging is available:

DCIO Merged PPMU Capabilities		
Number of Merged Channels	Total Supplies per Instrument	Maximum Merged Current
2	96	128mA
4	48	256mA
8	24	512mA

16	12	1.024A
32	4	2.048A

You must connect merged DCIO channels on the DIB. For information on designing DCIO DIBs that use the merged PPMU feature, see DCIO DIB Design Guidelines.

Merged channels must be adjacent by channel number. For example, when merging two channels, channel 0 must be merged with channel 1, channel 2 with channel 3, and so forth. In a four-channel configuration, channels 0-3 are merged, channels 4-7 are merged, and so forth.

In a 32-channel configuration, the last 16 channels of each 48-channel group are not available. Otherwise, any combination of channel merging is allowed. For example, a block of 48 channels can have one group of 16 merged channels, a group of eight merged channels, two groups of four merged channels, three groups of two merged channels, and 10 unmerged channels.

For information on how to program DCIO merged PPMUs, see Merged PPMU Operation.

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